Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Real-Time Audio, C++, DSP, Pattern Detection | Applied ML

Canadian Permanent Resident · Eligible to work in Canada without sponsorship

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Professional Summary

Machine Learning Engineer with 10+ years spanning across industry and academia. I build end-to-end ML systems: from data ingestion and feature engineering through model training, orchestration, deployment, and iteration. My work spans across real-time and streaming ML, time-series modeling, and pattern/anomaly detection across sequential data, computer vision, and structured datasets. Skilled at working with product, data, and engineering teams to translate ambiguous business problems into scalable, reliable AI solutions. Strong C++ and real-time audio background. Published researcher with production-grade results (<30ms real-time inference, >90% F1, and up to 300% efficiency improvement in deployed industrial solutions).

Core Competencies

ML Engineering: End-to-End ML Lifecycle, Production Deployment, Real-Time Audio, Real-Time / Low-Latency Inference, Performance Optimization, ML Orchestration, Multi-Tenant ML Systems, MLOps, Prompt engineering, AI Ethics, Human-in-the-Loop Systems
Domain Expertise: Time-Series Analysis, Anomaly Detection, Recommender Systems, Ranking Systems, Computer Vision, Multimodal Data Modeling, Embeddings, Pattern Recognition, NLP, Signal Processing / DSP, Generative AI
Research & Leadership: Benchmarking & Ablations, Experimental Design, Algorithm Design, Team Leadership, Cross-Functional Collaboration, Technical Writing, Peer-Reviewed Publications

Technical Stack

Languages: C++ (C++17), Python, C, JavaScript, MATLAB, C#
ML / AI: Tensorflow, Keras, PyTorch, JAX, scikit-learn, HuggingFace Transformers, LLMs, RAG, AWS SageMaker, Transfer Learning, OpenCV, Librosa, JUCE, VST, Elk Audio OS, NumPy, Pandas, SciPy
Data & Cloud: AWS (EC2, S3, ECS, MWAA, RDS), Snowflake, SQL, ETL Pipelines, Information Retrieval
Dev & MLOps: Docker, Kubernetes, MLFlow, Apache Airflow, Jenkins, Git, Linux/UNIX, CMake, Pytest, Unit Testing, CI/CD, REST APIs, ONNX

Professional Experience

Independent Researcher | *Open Source / MUSMET external collaborator, Toronto, ON* *Jan 2026 – present*

- Released Nebula, an open-source C++17 audio feature-extraction library with latency benchmarking on ARM.
- Actively researching on foundation model-based synthetic dataset curation for limited labeled data regimes.
- Authored 3 publications on embedded real-time pattern detection and edge-to-server distributed feature extraction.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 – Dec 2025*

- Owned full ML pipelines for streaming multi-dimensional data (EU-funded MUSMET project), covering data acquisition, signal processing, feature engineering, training/evaluation, and deployment with monitoring.
- Built real-time audio inference services (C++, JUCE VST plugin) hitting <30ms latency at >90% F1, with reproducible pipelines and documentation structured for long-term maintainability
- Designed benchmark suites (baselines + ablations) to quantify architecture/feature tradeoffs, improve generalization, and guide system-level decisions under compute constraints.
- Coordinated with academic and industry partners to move ML prototypes from research into documented, production-grade components, covering the full arc from novel prototyping to deployment.

Visiting Researcher | *McGill University, Montreal, Canada* *May 2024 – Aug 2024*

- Prototyped and evaluated a real-time gesture detection ML system on multimodal sensor + time-series data. Assessed deployment feasibility and compute trade-offs for near real-time inference.
- Explored generative AI methods (diffusion-based and VAE-based synthetic data generation) to improve model robustness under limited labeled data regimes, with findings directly shaping deployment decisions.

Visiting Researcher | *University of Visual and Performing Arts, Colombo 02, Sri Lanka* *Jan 2024 – Mar 2024*

- Ran human-centered evaluations of ML-driven interactive systems, combining performance metrics and user feedback to produce concrete recommendations on usability, interpretability, and adoption readiness.

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo 05, Sri Lanka* *Aug 2020 – Feb 2021*

- Designed and deployed ML and statistical modeling pipelines for financial forensics, compliance monitoring, and risk detection on large structured enterprise datasets.
- Worked with compliance and risk stakeholders to define detection requirements and turn them into technical specs.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 – Jul 2020*

- Led end-to-end development and deployment of ML and computer vision systems for manufacturing at scale, integrating camera + sensor streams into production workflows; improved operational efficiency by up to 300%.
- Built real-time data ingestion and ETL pipelines for high-frequency IoT and operational time-series data across distributed sites, with quality checks and monitoring to keep reliability consistent.
- Developed ML-driven optimization and decision engines to support real-time monitoring and long-term capacity/efficiency planning from streaming machine and sensor data.
- Deployed human-in-the-loop inspection models balancing throughput against operational risk; mentored engineers and established code review and CI/CD practices to accelerate ML adoption across teams.
- Worked with product, operations, and engineering stakeholders to define scope and milestones, turning business needs into measurable technical solutions.

Education

Ph.D - Information Engineering and Computer Science - *University of Trento, Italy* *Jan 2025*

M.Sc - Telecommunication and Electronic Engineering - *Sheffield Hallam University, UK* *Aug 2018*

B.Eng (Hons) - Electronic Engineering - *Sheffield Hallam University, UK* *Mar 2014*

Selected Publications (full list of publications available at: *nishal.xyz/publications*.)

- Silva, N. and Turchet, L. *Real-Time Audio Pattern Detection for Smart Musical Instruments*. Journal of the Audio Engineering Society, Mar. 2026.
- Silva, N., Wanderley, M. and Turchet, L. *Melody and Motion: Integrating Guitar Gesture Detection with Musical Patterns for Extended Control....* IEEE International Symposium on the Internet of Sounds, Oct. 2025.

Independent Research , Awards, Volunteer Work

- **MIDI Innovation Awards:** Finalist (Sep 2023): Real-time symbolic pattern detection system for smart instruments.
- **GMOA Recognition (Sri Lanka):** Computer vision system for COVID-19 patient vitals monitoring, deployed across hospital wards during the pandemic; enabled remote monitoring and reduced healthcare worker exposure.
- Built real-time ML prototypes for audio and sensor streams under strict latency limits, tested in live performances.